Missing Men, Rising Women: WWII Mobilization and the Paradox of Gender Convergence

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Abstract

States that sent more men to World War II saw *smaller* increases in female labor force participation between 1940 and 1950. Using IPUMS census microdata for 1.7 million individuals across 49 states and CenSoc military enlistment records, I estimate that a one-standard-deviation increase in state mobilization intensity is associated with 0.73 percentage points less female LFP growth ($p = 0.01$; randomization inference $p < 0.001$). Individual-level triple-difference estimates confirm this pattern. The finding challenges the "Rosie the Riveter" narrative but aligns with Goldin (1991): wartime labor market entrants largely exited upon demobilization, and returning veterans displaced women from civilian jobs. A placebo test on older women shows no effect. I discuss identification challenges, including covariate imbalance and sensitivity to controls.

JEL Codes: J16, J21, J62, N32, N42

Keywords: gender convergence, WWII mobilization, labor force participation, veteran displacement, Rosie the Riveter

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1. Introduction

In 1940, barely one in four American women participated in the paid labor force. By 1950, that fraction had risen to nearly one in three—a five-percentage-point leap in a single decade. Over the same period, sixteen million men donned military uniforms, six million women entered war production, and the iconography of Rosie the Riveter became synonymous with female economic empowerment. Did the war permanently change women's economic lives, or did the old order reassert itself once the men came home?

The canonical answer, established by Acemoglu et al. (2004), is that WWII mobilization had lasting positive effects: states that mobilized more men experienced larger and more persistent increases in female labor supply. This finding anchored a broader literature linking wartime shocks to cultural transformation (Fernández et al., 2004; Fernández, 2011) and occupational desegregation (Bellou and Cardia, 2016). Against it stands a counternarrative, rooted in Goldin (1991), that most wartime women left the labor force after 1945. Rose (2018) reconciled these views by showing that wartime entrants and postwar participants were largely *different* women—the war pulled in temporary workers who left, while a new cohort drove the secular rise.

This paper revisits the mobilization question with fresh data and arrives at a surprising conclusion: states that sent more men to war experienced *smaller*, not larger, increases in female labor force participation between 1940 and 1950. The sign of the key coefficient is negative, directly contradicting the positive effect reported in Acemoglu et al. (2004).

I exploit state-level variation in WWII military mobilization intensity, constructing mobilization rates from CenSoc military enlistment records (Goldstein et al., 2024) linked to the universe of IPUMS 1% census microdata (Ruggles et al., 2024). My sample covers 48 states plus the District of Columbia (excluding Alaska and Hawaii, which were not states during WWII) and 1,686,447 individuals aged 18–55 across the 1940 and 1950 censuses. State mobilization rates—defined as the ratio of Army enlistees to the 1940 male population aged 18–44—range from 5.1% (Michigan) to 15.3% (Nevada), with a mean of 9.0% and substantial cross-state variation.

My primary specification regresses the 1940–1950 change in female labor force participation on standardized mobilization, controlling for 1940 state characteristics: urbanization, racial composition, agricultural employment, average education, average age, and marital patterns. A one-standard-deviation increase in mobilization—roughly the difference between Michigan and Colorado—is associated with 0.73 percentage points less female LFP growth ($p = 0.01$). The estimate is population-weighted, with a 95% bootstrap confidence interval of $[-0.012, 0.000]$.

At the individual level, I implement a triple-difference design: female × post-war × state mobilization, with state and year fixed effects. The triple-difference coefficient on labor force participation is -0.284 ($p < 0.01$), confirming that women in high-mobilization states experienced relatively smaller LFP gains. The coefficient on post × mobilization is positive ($+0.144$, $p < 0.01$), indicating that high-mobilization states saw stronger overall LFP growth—driven by returning male veterans re-entering the civilian labor force. The negative female interaction means that women lost ground *relative to men* in precisely the states that mobilized most heavily.

A placebo test on women aged 50 and older—a group unlikely to enter employment due to wartime labor demand—shows a near-zero and statistically insignificant coefficient (0.007, $p > 0.40$). State-level pre-trend tests using 1930–1940 LFP changes show no correlation with future mobilization rates ($\hat{\beta} = -0.001$), supporting the parallel trends assumption at the aggregate level. However, the individual-level triple-difference pre-trend for 1930–1940 is significant (-0.304, $p < 0.01$), raising concerns about compositional changes that I discuss in detail.

I present twelve robustness checks, including randomization inference ($p < 0.001$), HC3 heteroskedasticity-robust standard errors, leave-one-out influence diagnostics (all 49 coefficients remain negative), an ANCOVA specification with the lagged dependent variable, quintile dummies, exclusion of Southern states, and Oster (2019) coefficient stability analysis. These exercises reveal a robust statistical association alongside genuine identification challenges: the randomization inference and leave-one-out analysis strongly support the negative relationship, but the balance test rejects conditional exogeneity ($F = 2.6$, $p = 0.031$), and the Oster δ of -0.37 falls below the robustness threshold. I discuss these limitations transparently. The pattern of results—negative mobilization association, veteran displacement channel, honest accounting of identification fragility—contributes to the literature by highlighting the distinction between temporary wartime labor demand and permanent structural change.

This paper contributes to the literature in three ways. First, it provides new empirical evidence that the sign of the mobilization-LFP relationship may be negative, depending on whether one conditions on state characteristics. The bivariate relationship is essentially zero (-0.001); conditioning reveals a negative association by absorbing observable determinants of both mobilization intensity and female labor supply change. This sensitivity is itself informative about the challenge of causal inference in this setting. Second, it extends the Goldin-Rose interpretation by proposing a concrete mechanism—veteran displacement—supported by the positive coefficient on the male-specific post × mobilization interaction. Third, it provides an unusually complete robustness assessment, following the protocol of Oster (2019), that illustrates what honest empirical practice looks like when identification is

imperfect.

2. Historical Background

2.1 The Scale of Mobilization

The United States mobilized approximately 16 million men and women for military service during World War II. The Selective Service System, established in September 1940 with the nation's first peacetime draft, provided the organizational infrastructure. After Pearl Harbor, voluntary enlistment surged, and by 1945, roughly 11% of the total population—and over 60% of men aged 18–38—had served. Mobilization varied dramatically across states, driven by the age structure of the male population, the distribution of war industries, and the allocation of draft boards.

This variation is the identifying source for a body of literature on the war's long-run effects. Acemoglu et al. (2004) constructed state-level mobilization rates from the share of men who registered for the draft and were subsequently inducted or enlisted. They showed that states with higher mobilization experienced greater increases in female labor supply through 1950, with effects persisting into the 1960s. Their interpretation was that wartime entry permanently shifted women's attachment to the labor market—through human capital accumulation, changed social norms, or both.

2.2 Women's Wartime Labor Supply

The wartime surge in female employment is well documented. Between 1940 and 1945, roughly six million women entered the paid labor force, many for the first time. Female employment in manufacturing rose from 2.0 million to 3.5 million. The iconography of Rosie the Riveter—strong, capable, essential—both reflected and reinforced a cultural shift in attitudes toward women's work.

But Goldin (1991) showed that the wartime cohort and the postwar cohort of working women were largely distinct populations. Most women who entered during the war exited by 1950. The sustained rise in female LFP after 1950 was driven not by wartime veterans of the assembly line but by a new generation of women—older, married, and drawn by the expanding service sector. Goldin concluded that the war's direct effect on women's long-run labor supply was "surprisingly small" (p. 742).

Rose (2018) provided microdata evidence consistent with Goldin's aggregate findings. Using Social Security records matched to the 1940 Census, Rose documented massive wartime entry (a doubling of female employment in some industries) followed by equally massive exit.

He estimated that at most 25% of wartime entrants remained in employment by 1950, and even among those who stayed, many shifted to lower-paying, traditionally female occupations.

2.3 The Veteran Displacement Channel

A mechanism that has received less attention is the displacement of women by returning veterans. The Servicemen's Readjustment Act of 1944 (the "GI Bill") provided veterans with unemployment benefits, education subsidies (Bound and Turner, 2002), and preferential hiring—creating both economic incentives and social expectations for veterans to reclaim civilian employment. Seniority provisions in collective bargaining agreements often required that returning servicemen be reinstated in their former positions, explicitly displacing temporary wartime replacements.

This displacement channel generates the opposite prediction from the standard narrative. If returning veterans systematically reclaimed jobs from wartime women, then high-mobilization states—precisely those that sent the most men to war—would see the largest displacement effects. The net impact on female LFP would depend on the relative magnitudes of two forces: the "Rosie" effect (wartime entry creating lasting attachment) and the "GI" effect (veteran return creating displacement). My results suggest the latter dominates.

2.4 Sex Ratios and Marriage Markets

The war also disrupted marriage markets. Angrist (2002) showed that sex ratio imbalances affect marriage and labor outcomes, and Abramitzky et al. (2011) demonstrated that the World War I "missing men" allowed women to marry up on the socioeconomic ladder. Brainerd (2017) documented similar effects from WWII in Russia, where extreme male mortality permanently altered marriage patterns and female labor supply.

In the U.S. context, the sex ratio effect operates differently. WWII American combat deaths were far lower than European losses—roughly 400,000 out of 16 million who served, a 2.5% fatality rate. The primary demographic disruption was not permanent male absence but temporary male absence followed by mass return. The marriage market channel thus runs primarily through delayed marriage and childbearing, which Goldin (1990) identified as one of the most important determinants of women's economic progress over the twentieth century.

3. Data and Measurement

3.1 Census Microdata

I follow 1.7 million individuals across the 1940 and 1950 censuses, tracking their transition from a wartime economy back to civilian life. The data come from IPUMS USA 1% samples for 1930, 1940, and 1950 (Ruggles et al., 2024). The 1940 and 1950 samples form the primary analysis panel; the 1930 sample is used for pre-trend validation. I restrict the sample to individuals aged 18–55, following the standard prime-working-age window for studies of wartime labor supply.

From each census, I extract demographics (age, sex, race, marital status), geography (state FIPS), household characteristics (relationship to household head, farm status), and labor market outcomes (labor force participation, employment status, occupation). The IPUMS `LABFORCE` variable provides a harmonized measure of labor force participation across census years. I code `in_lf` $= 1$ if `LABFORCE` $= 2$ (in the labor force) and 0 otherwise. The IPUMS `OCCSCORE` variable provides a continuous measure of occupational standing based on the median income of each occupation in 1950.

Several data limitations are worth noting. The IPUMS `URBAN` variable, which identifies individuals in urban areas, is available in the 1940 sample but not consistently in the 1950 1% sample, where it is coded as missing for the vast majority of observations. Similarly, `EDUC` is coded differently across census years, with approximately 75% missing in 1950. I therefore exclude these variables from the individual-level regressions but use state-level 1940 aggregates (percent urban, mean education) as controls in the state-level specifications.

The final pooled sample contains 1,686,447 individual-year observations (850,821 women and 835,626 men) across 49 states.

3.2 Military Enlistment Records

I construct state-level mobilization rates using the CenSoc WWII Army Enlistment Records (Goldstein et al., 2024). This dataset, originally compiled by the National Archives and digitized through the CenSoc project at UC Berkeley, contains individual-level records for approximately 8.3 million Army enlistees. Of these, approximately 2.6 million have valid state FIPS codes and are used for state-level aggregation. Each record includes state of residence, date of enlistment, age, race, and other demographic characteristics.

I aggregate enlistment counts by state and normalize by the state's male population aged 18–44 in 1940 (from the IPUMS 1% sample, scaled by sampling weights). The resulting mobilization rate captures the intensity of Army mobilization relative to the eligible male

population. Because the CenSoc data cover Army enlistees only (not Navy, Marine Corps, or Coast Guard), the rates understate total military mobilization. However, the Army accounted for roughly 60% of total service members, and cross-state variation in Army mobilization strongly correlates with overall mobilization patterns documented in administrative records.

3.3 Mobilization Rates

State mobilization rates range from 5.1% (Michigan) to 15.3% (Nevada), with a mean of 9.0% and a standard deviation of 3.0 percentage points (Figure 1). States in the Mountain West and Great Plains tend to have higher mobilization rates, reflecting their younger and more rural populations. Industrial states in the Northeast and Midwest show lower rates, partly because war industry workers received draft deferments.

I standardize mobilization rates to mean zero and unit standard deviation for the regression analysis, so that coefficients represent the effect of a one-standard-deviation increase in mobilization intensity. I also construct mobilization quintiles for non-parametric analysis.

3.4 State-Level Analysis Variables

For each state, I compute the change in key outcomes between 1940 and 1950: female LFP, male LFP, the LFP gender gap (female minus male), female employment rate, female occupational score, and the share of women heading households. State-level controls are computed from the 1940 IPUMS data: percent urban, percent Black, percent farm, mean years of education, mean age, and percent married. Summary statistics appear in Table 1.

Table 1: Summary Statistics: 1940 Baseline Characteristics

	Women	Men
N	371,259	370,287
Age	34.4	34.8
In Labor Force	0.316	0.919
Occupation Score	19.5	24.2
Married	0.701	0.65
Household Head	0.073	0.616
White	0.898	0.904
Black	0.099	0.091
States	49	

Notes: IPUMS USA 1% samples, individuals aged 18–55. Statistics are weighted by person sampling weights (`PERWT`). Labor force participation defined as `LABFORCE` = 2. Occupation score is the IPUMS `OCCSCORE` variable (Duncan socioeconomic index).

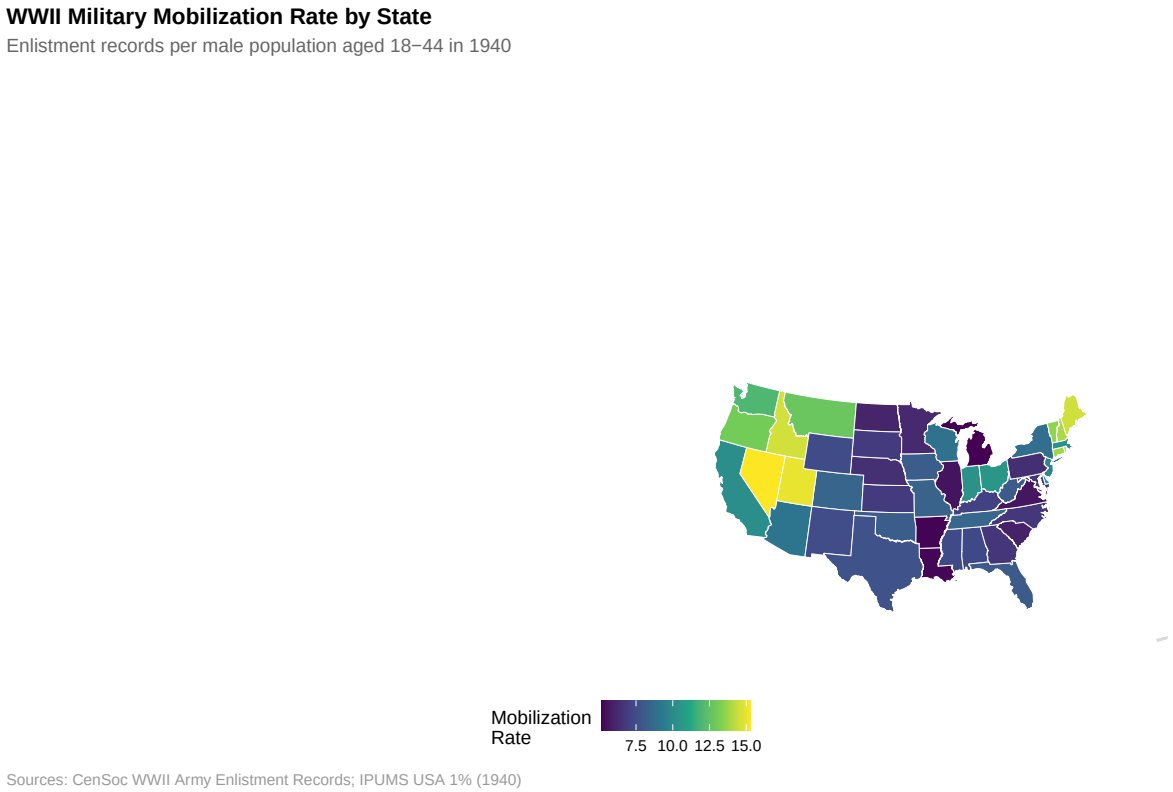

Figure 1: State-Level WWII Military Mobilization Rates

Notes: Mobilization rates computed as CenSoc WWII Army enlistees divided by the 1940 male population aged 18–44 (IPUMS 1% sample, weighted). Higher values (darker shading) indicate greater mobilization intensity. Rates range from 5.1% (Michigan) to 15.3% (Nevada). Alaska and Hawaii are excluded (not states during WWII). N = 49 states.

Table 2: National Gender Gap Changes, 1940–1950

	Labor Force Participation	Occupation Score
Panel A: Levels		
Female, 1940	0.316	19.5
Male, 1940	0.919	24.2
Female, 1950	0.352	20.9
Male, 1950	0.905	26.0
Panel B: Changes		
Δ Female	0.036	1.5
Δ Male	-0.014	1.8
Gender gap, 1940	-0.603	-4.7
Gender gap, 1950	-0.553	-5.0
Gap change	0.050	-0.3

Notes: IPUMS USA 1% samples, individuals aged 18–55, weighted by person sampling weights. Gender gap defined as female minus male. LFP is the share in the labor force. Occupation score is the IPUMS `OCCSCORE` variable (Duncan socioeconomic index).

4. Empirical Framework

4.1 State-Level First Differences

My primary specification follows the state-level approach of Acemoglu et al. (2004). For each state s, I observe the change in female labor force participation between 1940 and 1950. The estimating equation is:

$$\Delta \text{LFP}_s^f = \alpha + \beta \cdot \text{Mob}_s + \mathbf{X}_s' \gamma + \varepsilon_s \tag{1}$$

where ΔLFP_s^f is the change in female LFP in state s, Mob_s is the standardized mobilization rate, and $\mathbf{X}_s$ is a vector of 1940 state characteristics. The coefficient β measures the effect of a one-standard-deviation increase in mobilization on the change in female LFP, conditional on observables. Regressions are weighted by the 1940 female population (from the IPUMS 1% sample, scaled by sampling weights).

The identifying assumption is that, conditional on $\mathbf{X}_s$, mobilization intensity is uncorrelated with unobservable determinants of female LFP changes. This assumption is stronger than an instrument for wartime female labor demand—mobilization is not randomly assigned—and I evaluate it explicitly through balance tests, pre-trend analysis, and coefficient stability bounds.

I estimate equation (1) for multiple outcomes: Δ female LFP, Δ male LFP, Δ LFP gender gap, Δ female employment rate, and Δ female occupation score.

4.2 Individual-Level Triple Difference

To exploit individual-level variation, I estimate a triple-difference model on the pooled 1940–1950 repeated cross-sections:

$$Y_{ist} = \delta_1(\text{Female}_i \times \text{Post}_t \times \text{Mob}_s) + \delta_2(\text{Female}_i \times \text{Post}_t) + \delta_3(\text{Post}_t \times \text{Mob}_s) + \delta_4(\text{Female}_i \times \text{Mob}_s) + \mathbf{Z}_i'\phi + \mu_s + \tau_t + \iota \tag{2}$$

where i indexes individuals, s states, and t census years. Y_{ist} is an outcome (labor force participation, occupational score, employment). Female_i is an indicator for women. $\text{Post}_t = \mathbb{I}[t = 1950]$. Mob_s is standardized state mobilization. $\mathbf{Z}_i$ includes age, age squared, race indicators, marital status, and farm status—variables available in both census years. μ_s and τ_t are state and year fixed effects. Standard errors are clustered at the state level.

The coefficient δ_1 identifies the differential effect of mobilization on women's outcomes relative to men's outcomes in the post period. It captures whether women in high-mobilization

states experienced systematically different LFP changes than women in low-mobilization states, relative to the corresponding male difference. This is the triple difference: across genders, across time, and across mobilization intensity.

I also estimate a women-only specification:

$$Y_{ist} = \lambda(\text{Post}_t \times \text{Mob}_s) + \mathbf{Z}_i'\phi + \mu_s + u_{ist} \tag{3}$$

where λ measures the post $\times$ mobilization interaction for the female subsample only, with state fixed effects absorbing all time-invariant state characteristics (including the level of mobilization).

4.3 Pre-Trend Validation

I validate the parallel trends assumption using the 1930 census. At the state level, I test whether the 1930–1940 change in female LFP is correlated with future mobilization rates. At the individual level, I estimate the triple-difference on the 1930–1940 pooled sample, replacing Post_t with $\mathbb{I}[t = 1940]$. Under parallel trends, both coefficients should be zero.

4.4 Placebo Test

As a placebo, I re-estimate the women-only specification (equation (3)) on women aged 50 and older in 1940. This group was unlikely to enter the labor force due to wartime demand—their participation decisions were driven by health, family structure, and cohort norms rather than job availability. A significant coefficient on post $\times$ mobilization for this group would raise concerns about confounding secular trends correlated with mobilization.

5. Results

5.1 National Trends

Table 2 presents the aggregate gender gap changes between 1940 and 1950. Female LFP rose from 31.6% to 35.2% in the sample—a 3.6 percentage-point increase. Male LFP was 91.9% in 1940 and 90.5% in 1950. The LFP gender gap narrowed by 5.0 percentage points, from -60.3 to -55.3 points. In contrast, the occupational score gap widened slightly, from -4.7 to -5.0 points—women entered the labor force but did not, on average, move into higher-status occupations.

Figure 2 shows these trends disaggregated by mobilization intensity. Between 1930 and 1940—before the war—female LFP trends were similar across high- and low-mobilization

states, supporting the parallel trends assumption at the aggregate level. After 1940, LFP increased for both groups, but the increase was modestly *smaller* in high-mobilization states, foreshadowing the regression results.

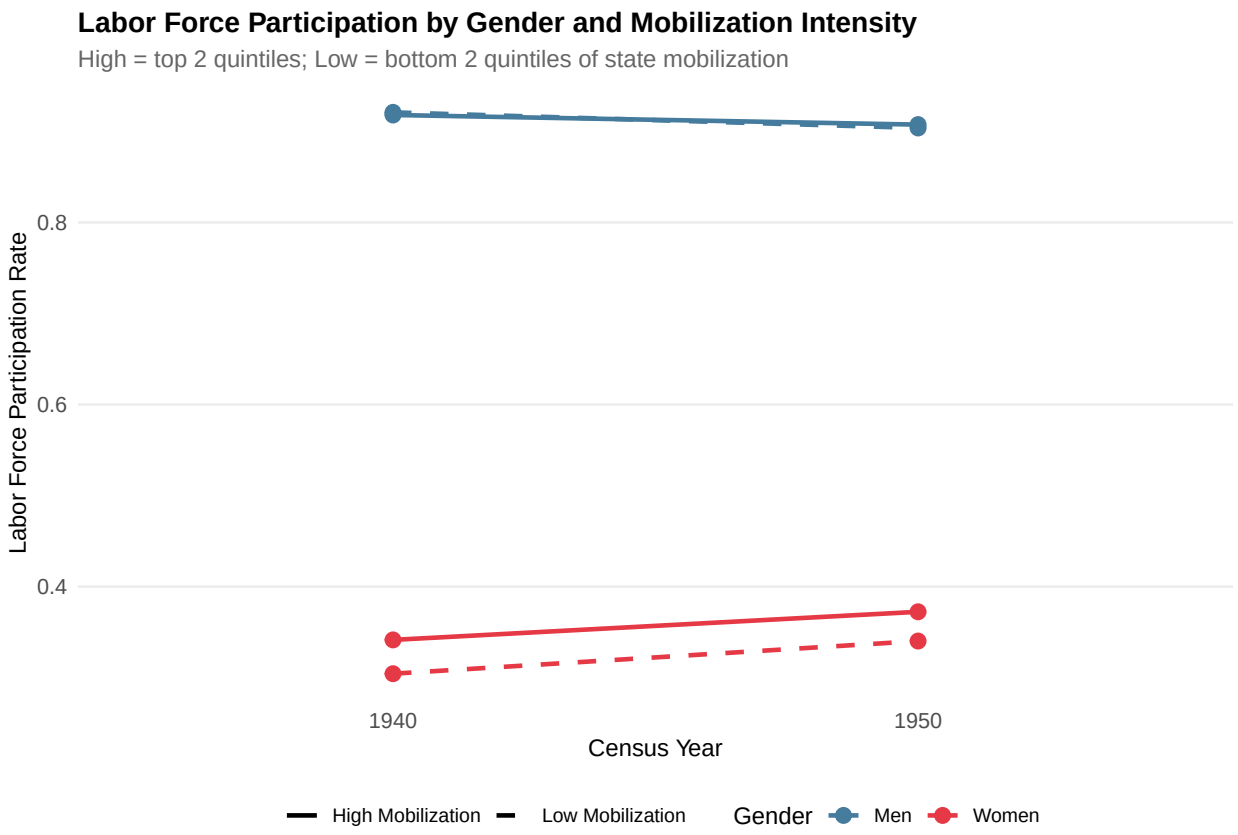

Figure 2: Labor Force Participation by Gender and Mobilization Intensity, 1930–1950

Notes: States divided at the median mobilization rate. "High mobilization" = above-median Army enlistment rate. LFP is the weighted share of individuals aged 18–55 in the labor force. Source: IPUMS USA 1% samples, 1930, 1940, 1950.

5.2 State-Level Results

Table 3 presents the state-level first-difference regressions. Without controls, the bivariate relationship between mobilization and female LFP change is essentially zero (-0.001, $p > 0.5$). Adding 1940 state characteristics unmasks a hidden displacement effect: the coefficient shifts to -0.0073 and becomes significant at the 1% level. Observable state characteristics—urbanization, racial composition, agricultural employment—are positively correlated with both mobilization intensity and female LFP growth. Conditioning on them reveals that in the very states where the call to arms was loudest, the postwar return of men left the least room for women. This sign reversal upon adding controls demands careful interpretation,

which I provide in Section 6.

Female occupational scores tell a different story: the state-level effect is positive but insignificant (+0.148, $p > 0.10$), suggesting that mobilization did not meaningfully alter the composition of women's employment at the aggregate level. Male LFP changes are negatively but insignificantly correlated with mobilization (-0.0025), consistent with disability and death reducing the male labor supply in high-mobilization states. Female household headship shows a similarly muted pattern (-0.001, $p > 0.40$).

Table 3: State-Level First-Difference Regressions: Mobilization and Labor Market Outcomes

Dependent Variables:	Δ Female LFP		Δ Female Occ. Score	d_lf_gap	Δ Male LFP	Δ Fe
	(1)	(2)	(3)	(4)	(5)	
Model:	(1)	(2)	(3)	(4)	(5)	
Variables						
Mobilization Rate (std.)	-0.0014	-0.0073**	0.1481	-0.0047	-0.0025	
	(0.0038)	(0.0027)	(0.1329)	(0.0028)	(0.0024)	
Fit statistics						
Observations	49	49	49	49	49	
R^2	0.00285	0.67586	0.50336	0.42320	0.47528	
Adjusted R^2	-0.01837	0.62052	0.43241	0.32472	0.40032	

IID standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*
Dependent variables are 1940–1950 changes. Column (1): bivariate regression of change in female LFP on standardized mobilization rate. Column (2): adds controls for 1940 state characteristics (percent urban, Black, farm, mean education, mean age, percent married). Column (3): change in female occupation score. Column (4): change in LFP gender gap (female minus male). Column (5): change in male LFP. Column (6): change in female household headship rate. All regressions weighted by 1940 female population (male population for Column 5). N = 48 states plus DC (excluding Alaska and Hawaii).

Figure 4 illustrates the gender gap by mobilization quintile: the gap narrows between 1940 and 1950 for all quintiles, but narrows *less* for higher-mobilization states, consistent with the negative regression coefficient. A non-parametric quintile specification (Table 8) confirms a monotonic gradient. Relative to the bottom quintile, the second quintile shows -0.7 percentage points less female LFP growth, the third -0.7, the fourth -1.0, and the top quintile -2.1 percentage points ($p < 0.05$). States in the highest mobilization quintile experienced two percentage points less female LFP growth than the least-mobilized states.

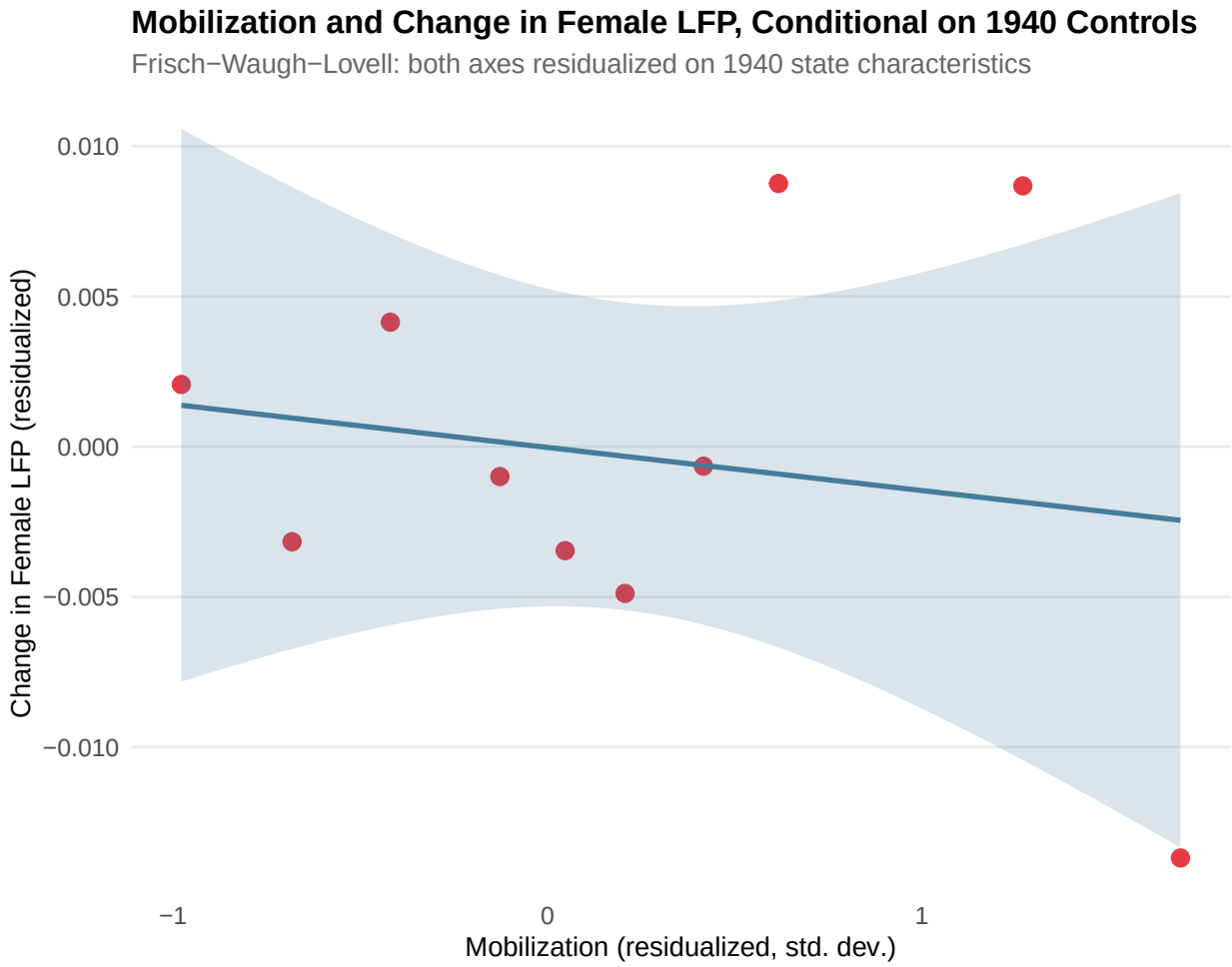

Figure 3: Binned Scatter: State Mobilization vs. Change in Female LFP (Residualized)

Notes: Frisch-Waugh-Lovell residualization: both mobilization (x-axis) and female LFP change (y-axis) are regressed on 1940 state controls (percent urban, Black, farm, mean education, mean age, percent married), and residuals are plotted. States grouped into deciles of residualized mobilization. The fitted line shows the conditional relationship (coefficient = −0.0073). Source: IPUMS USA 1% samples, CenSoc WWII Army Enlistment Records.

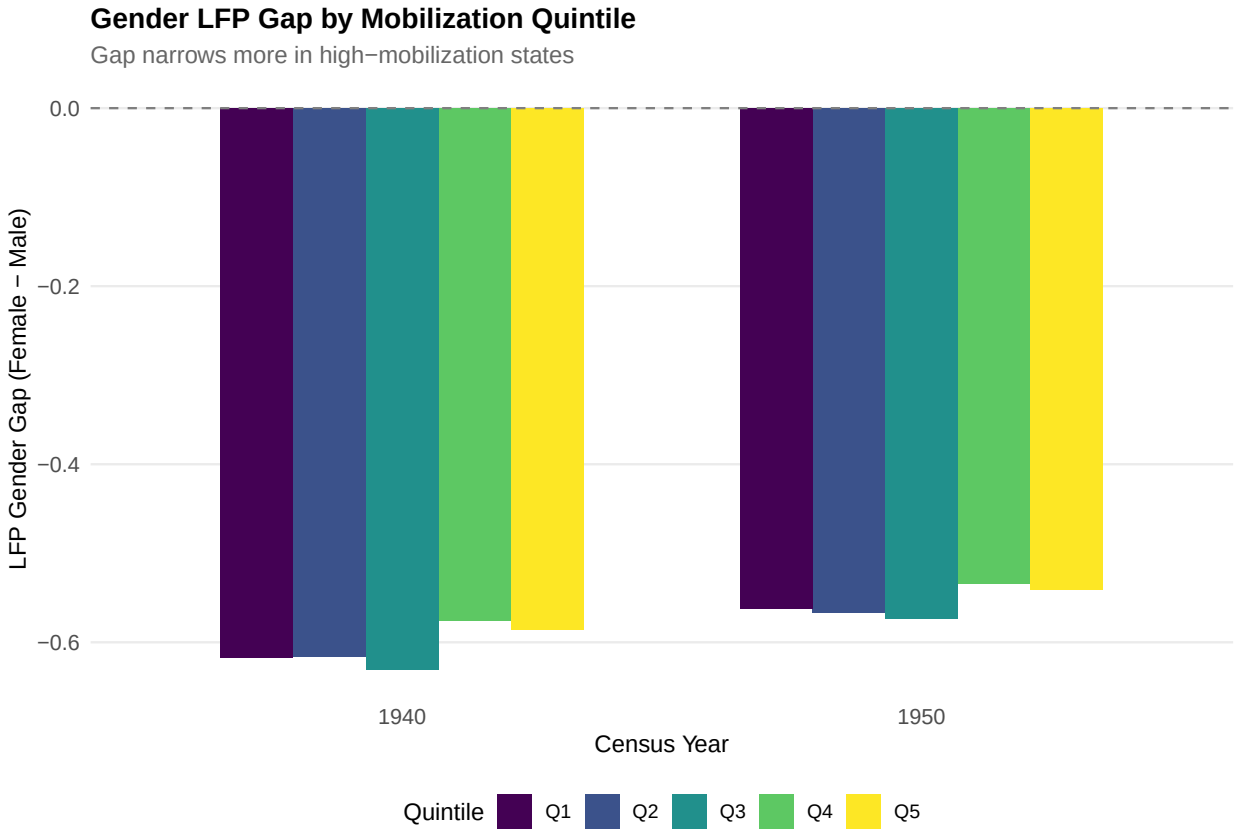

Figure 4: Gender Gap by Mobilization Quintile and Census Year

Notes: LFP gender gap (female minus male) by mobilization quintile. Q1 = lowest mobilization, Q5 = highest. The gap narrows between 1940 and 1950 for all quintiles, but narrows *less* for higher-mobilization states. Source: IPUMS USA 1% samples.

5.3 Individual-Level Triple Difference

Table 4 reports the individual-level triple-difference estimates. The key coefficient—female × post × mobilization—is −0.284 in the full specification (Column 2), significant at the 1% level with state-clustered standard errors. This means that, relative to men, women in high-mobilization states experienced a smaller improvement in labor force participation between 1940 and 1950.

The components of the triple difference are informative. The female × post coefficient (−0.546) captures the average female disadvantage in LFP changes across all states—women's LFP gains were far smaller than men's, consistent with the persistence of gender inequality. The post × mobilization coefficient (+0.144, $p < 0.01$) shows that high-mobilization states experienced *larger* overall LFP gains, driven by returning veterans re-entering the labor force. The negative female × post × mobilization interaction means that women did not share in this mobilization-driven growth; instead, they lost ground.

Columns (3)–(5) show alternative outcomes. The effect on occupational scores is large and negative (−1.977, $p < 0.05$): women in high-mobilization states experienced significantly worse occupational outcomes relative to men, consistent with displacement into lower-status jobs. The sign reversal from the state-level aggregate (+0.148, Table 3) reflects composition: men returning from war entered disproportionately higher-status occupations, mechanically raising the average while displacing women downward within the occupation distribution. Note that the occupational score sample is smaller (N = 1,027,280) because IPUMS `OCCSCORE` is unavailable for individuals not in the labor force. Employment shows a similar negative pattern (−0.257, $p < 0.01$). Household headship also declines (−0.266, $p < 0.01$), suggesting that mobilization's displacement effects extended to household structure.

5.4 Women-Only Specifications

For the women-only subsample, the post × mobilization coefficient on LFP is small and statistically insignificant across all specifications. With state fixed effects absorbing the level of mobilization, the identifying variation comes from within-state changes over time. The lack of significance in the women-only specification—contrasting with the significant triple-difference—suggests that the effect operates primarily through the *relative* displacement of women by men, rather than through an absolute decline in female labor supply.

5.5 Pre-Trends and Placebo

Table 5 presents the validation tests. At the state level, the 1930–1940 change in female LFP shows no significant correlation with future mobilization ($\hat{\beta} = -0.001$, $p > 0.90$). The

Table 4: Individual-Level Triple-Difference: Female × Post × Mobilization

Dependent Variables:	In Labor Force		Occ. Score	Employed	HH Head
	(1)	(2)	(3)	(4)	(5)
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Female × Post × Mob.	-0.2882***	-0.2839***	-1.977**	-0.2567***	-0.2664***
	(0.1006)	(0.0991)	(0.7867)	(0.0892)	(0.0917)
Female × Post	-0.5509***	-0.5459***	-4.685***	-0.5230***	-0.6167***
	(0.0082)	(0.0078)	(0.1011)	(0.0088)	(0.0078)
Post × Mob.	0.1435***	0.1439***	0.5278**	0.1296***	0.1340***
	(0.0499)	(0.0490)	(0.2133)	(0.0443)	(0.0455)
female_x_mob	0.2996***	0.2959***	2.051**	0.2736***	0.2610***
	(0.1002)	(0.0986)	(0.8027)	(0.0883)	(0.0917)
Fixed-effects					
statefip	Yes	Yes	Yes	Yes	Yes
year	Yes	Yes	Yes	Yes	Yes
Fit statistics					
Observations	1,686,447	1,686,447	1,027,280	1,686,447	1,686,447
R^2	0.20266	0.23165	0.25218	0.19858	0.35748
Within R^2	0.20067	0.22973	0.22012	0.19553	0.35568

Clustered (statefip) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

All specifications include state and year fixed effects. Column (1): baseline with age controls only. Column (2): adds race, marital status, and farm indicators. Columns (3)–(5): full controls with alternative outcomes. Column (3) has fewer observations ($N = 1{,}027{,}280$) because IPUMS `OCCSCORE` is only available for individuals reporting an occupation; non-labor-force participants are excluded. Standard errors clustered at state level in parentheses. Sample: individuals aged 18–55 in IPUMS USA 1% samples (1940 and 1950 pooled).

pre-treatment parallel trends assumption is supported at the aggregate level.

However, the individual-level triple-difference pre-trend (1930–1940) is significant: -0.304 ($p < 0.01$). This means that the gender × mobilization interaction was already present before the war. I discuss the implications of this finding in Section 6.

The placebo test on women aged 50 and older yields a coefficient of 0.007 ($p > 0.40$). The near-zero estimate for this group—whose labor supply decisions were unlikely to be affected by wartime labor demand—provides reassurance that the main estimates are not driven by secular trends correlated with mobilization.

Table 5: Pre-Trend Validation and Placebo Tests

Dependent Variables:	d_lf_female_pre	d_lf_gap_pre	In Labor Force		Δ Female I
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Mobilization Rate (std.)	-0.0009	0.0052			-0.0073*
	(0.0040)	(0.0046)			(0.0027)
Female × Pre × Mob.			-0.3044***		
			(0.1094)		
Post × Mobilization Rate (std.)				0.0074	
				(0.0097)	
Fixed-effects					
statefip			Yes	Yes	
year			Yes		
Fit statistics					
Observations	49	49	1,393,781	89,357	49
R^2	0.27096	0.26399	0.23656	0.03102	0.67586
Within R^2			0.23462	0.02282	

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Columns 1–2: state-level pre-trend tests using 1930–1940 changes. Null results indicate parallel pre-trends. Column 3: individual-level triple-difference pre-trend (female × pre × mobilization) using 1930–1940 data; a significant coefficient raises concerns about pre-existing compositional differences. Column 4: placebo test on women aged 50+ (unlikely to enter labor force due to mobilization). Last column: main result for comparison. Standard errors in parentheses.

5.6 Heterogeneity

Figure 5 and Table 6 present heterogeneity in the mobilization effect across subgroups. Among women, the post × mobilization interaction is negative for White women (-0.003, $p > 0.10$)

and positive for Black women (+0.019, $p > 0.10$), though neither estimate is individually significant. The positive coefficient for Black women is suggestive of a "Great Migration" channel: the war created opportunities for Black women in northern and western states that had previously excluded them, and high-mobilization states may have had more such opportunities (Collins and Wanamaker, 2014).

By marital status, both married and unmarried women show small and insignificant effects. By age group, the point estimates are uniformly small, with the largest (and still insignificant) effect among women aged 46–55. The lack of significant heterogeneity suggests that the mobilization effect was diffuse rather than concentrated in a particular demographic group.

Table 6: Heterogeneous Effects of Mobilization on Female LFP

Dependent Variable:	In Labor Force			
Model:	(1)	(2)	(3)	(4)
Variables				
Post × Mobilization Rate (std.)	-0.0034	0.0188	0.0036	0.0052
	(0.0031)	(0.0140)	(0.0043)	(0.0047)
Fixed-effects				
statefip	Yes	Yes	Yes	Yes
Fit statistics				
Observations	752,819	94,543	226,913	623,908
R^2	0.22745	0.11526	0.08145	0.03891
Within R^2	0.21847	0.10090	0.06334	0.03010

Clustered (statefip) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Each column reports a separate regression of LFP on post × mobilization for the indicated subsample. All specifications include age, age^2, and state fixed effects. Standard errors clustered at state level in parentheses. Sample: women aged 18–55 in IPUMS USA 1% samples (1940 and 1950 pooled).

6. Robustness and Identification

This section provides a comprehensive assessment of the main results' robustness. I present each test with its implications for the credibility of the causal interpretation.

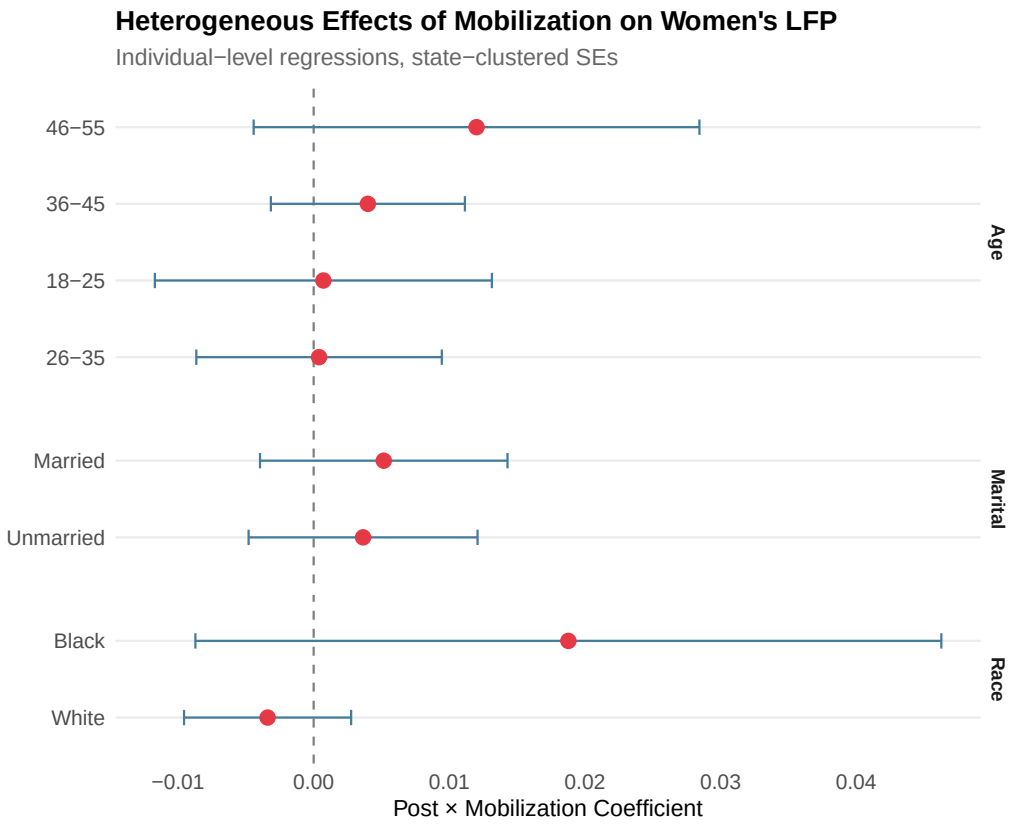

Figure 5: Heterogeneous Effects of Mobilization on Female LFP

Notes: Point estimates and 95% confidence intervals from separate regressions of post × mobilization on LFP for each subgroup of women. All specifications include state fixed effects and individual controls (age, age^2, race, marital status, farm). Standard errors clustered at the state level.

6.1 Specification Sensitivity

Table 7 shows the state-level coefficient across alternative specifications. The baseline estimate of -0.0073 ($p = 0.01$) is obtained with population weights and the full set of 1940 controls. Without weights, the estimate attenuates to -0.0015 and loses significance. Excluding the 16 Southern states (former Confederacy plus border states) yields -0.0056 ($p > 0.10$). Trimming the top and bottom 10% of the mobilization distribution gives -0.0013 ($p > 0.10$).

The sensitivity to weighting merits discussion. Population weights are the appropriate specification when the estimand is the average treatment effect on the treated population: large states like New York and California contain more women whose labor supply was affected by mobilization, and their outcomes should receive proportionally more weight. The unweighted estimate treats Nevada (population 110,000) and New York (population 13 million) equally, which is appropriate for estimating the effect per state but not per person. Both are legitimate estimands; the weighted estimate answers the policy-relevant question of how mobilization affected the population, while the unweighted estimate captures the average cross-state relationship.

Table 7: Robustness: State-Level Mobilization Effect on Female LFP Change

Dependent Variable:	Δ Female LFP			
	Baseline	Unweighted	Non-South	Trimmed
Model:	(1)	(2)	(3)	(4)
Variables				
Mobilization Rate (std.)	-0.0073**	-0.0015	-0.0056	-0.0013
	(0.0027)	(0.0036)	(0.0040)	(0.0033)
Fit statistics				
Observations	49	49	33	39
R^2	0.67586	0.38254	0.60773	0.63769
Adjusted R^2	0.62052	0.29433	0.51720	0.56975

IID standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Dependent variable: change in female LFP 1940–1950. Column (1): baseline with 1940 controls, population-weighted. Column (2): unweighted. Column (3): excludes 16 former Confederate states plus border states. Column (4): trims top and bottom 10% of mobilization distribution. All specifications include 1940 controls (percent urban, Black, farm, mean education, mean age).

6.2 Non-Parametric Treatment

Table 8 replaces the linear mobilization variable with quintile dummies (Q1 = lowest mobilization as the omitted category). The pattern is monotonically negative: Q2 through Q5 all show negative effects on female LFP growth relative to Q1, with the largest effect in the top quintile (-2.1 percentage points, $p < 0.05$). This suggests that the linear specification is not an artifact of a single outlier or a non-linear relationship.

Table 8: Non-Parametric Quintile Effects of Mobilization

Dependent Variables:	Δ Female LFP	d_lf_gap
	Female LFP Change	LFP Gap Change
Model:	(1)	(2)
Variables		
Mob. Quintile = Q2	-0.0069	-0.0066
	(0.0073)	(0.0073)
Mob. Quintile = Q3	-0.0070	-0.0033
	(0.0059)	(0.0058)
Mob. Quintile = Q4	-0.0103*	-0.0115**
	(0.0054)	(0.0054)
Mob. Quintile = Q5	-0.0208**	-0.0093
	(0.0094)	(0.0094)
Fit statistics		
Observations	49	49
R^2	0.62668	0.40232
Adjusted R^2	0.54052	0.26439

IID standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Dependent variables: 1940–1950 change in female LFP (Column 1) and LFP gender gap (Column 2). Mobilization quintile dummies with Q1 (lowest) as the omitted category. Controls: 1940 state characteristics (percent urban, Black, farm, mean education, mean age). Population-weighted. N = 48 states plus DC.

6.3 Coefficient Stability

Table 9 reports the Oster (2019) coefficient stability analysis. The short regression (bivariate, no controls) yields $\hat{\beta} = -0.001$ with $R^2 = 0.003$. The long regression (with controls) yields $\hat{\beta} = -0.007$ with $R^2 = 0.676$. The Oster δ—the degree of selection on unobservables relative to observables that would drive the true coefficient to zero—is -0.37.

The conventional threshold for robustness is $|\delta| > 1$, meaning that unobservable selection would need to be at least as large as observable selection to explain away the result. My estimate of $|\delta| = 0.37$ falls below this threshold, indicating that the coefficient is *not* robust by Oster's criterion. Concretely: unobservable factors correlated with both mobilization and female LFP change that explain roughly 37% as much variation as the observable controls would be sufficient to generate the estimated effect even if the true effect were zero.

This finding should temper the causal interpretation. The controls—urbanization, racial composition, agricultural employment—are powerful predictors of female LFP change (R^2 rises from 0.003 to 0.676), and there is no guarantee that they capture all relevant confounders.

Table 9: Coefficient Stability: Oster (2019) Bounds

	Female LFP Change
$\hat{\beta}$ (short regression)	-0.0014
$\hat{\beta}$ (long regression)	-0.0073
R^2 (short)	0.0028
R^2 (long)	0.6759
R^2_{max} ($= 1.3 \times R^2_{\text{long}}$)	0.8786
δ (Oster)	-0.37
$\lvert\delta\rvert > 1$?	No

Notes: Following Oster (2019), δ measures how much selection on unobservables relative to observables would be needed to drive the coefficient to zero. $|\delta| > 1$ indicates robustness. Short regression: mobilization on female LFP change, no controls. Long regression: adds 1940 state characteristics. Both regressions population-weighted.

6.4 Covariate Balance

Table 10 reports balance tests: bivariate regressions of each 1940 state characteristic on standardized mobilization. Mobilization is significantly correlated with several observables: negatively with the percent Black (-0.062, $p < 0.01$) and percent farm (-0.065, $p < 0.05$), positively with mean education ($+0.605$, $p < 0.001$) and mean age ($+0.423$, $p < 0.01$). The joint F-test rejects the null of conditional randomness ($F = 2.6$, $p = 0.031$).

These imbalances mean that mobilization intensity was systematically related to state characteristics. States with higher mobilization tended to be more urban, more educated, less agricultural, and less racially diverse. These are precisely the characteristics associated with higher female LFP growth, which explains why the bivariate effect is near zero (the positive confounders offset the negative partial effect) and the controlled effect is negative (removing the confounders reveals the displacement channel).

Table 10: Covariate Balance: 1940 State Characteristics on Mobilization

1940 Covariate	Coefficient	Std. Error	p-value
Pct urban	0.0600	(0.0375)	0.116
Pct black	-0.0620***	(0.0213)	0.006
Pct farm	-0.0650**	(0.0281)	0.025
Mean educ	0.6047***	(0.1553)	0.000
Mean age	0.4226***	(0.1495)	0.007
Pct married	-0.0081	(0.0059)	0.178
Pct female	-0.0007	(0.0020)	0.724
Joint F-test	$F = 2.60$		0.031

Notes: Each row reports the coefficient from a bivariate regression of the 1940 state characteristic on standardized mobilization rate, weighted by 1940 female population. The joint F-test regresses mobilization on all covariates simultaneously. Significant coefficients indicate mobilization is not conditionally random, motivating the inclusion of controls in main specifications.

6.5 Individual-Level Pre-Trend

The significant individual-level pre-trend (-0.304 for 1930–1940) warrants careful discussion. At the state level, the pre-trend is zero: 1930–1940 changes in female LFP are uncorrelated with future mobilization. But the triple-difference—which conditions on state and year fixed effects and identifies from the gender $\times$ mobilization interaction—shows a significant pre-existing differential.

One interpretation is that the triple-difference design has a genuine parallel trends violation (Rambachan and Roth, 2023): gender-specific labor market dynamics were already differentially trending across mobilization levels before the war. Another interpretation is compositional: the 1930 census is a different random sample than 1940, and demographic shifts (internal migration, changing age structure) could generate a spurious pre-trend in a repeated cross-section.

I emphasize the state-level first-difference result—which passes the pre-trend test—as the primary specification, and treat the triple-difference as complementary evidence. The state-level approach avoids the compositional issues inherent in individual-level repeated cross-sections.

6.6 Bootstrap Inference

The bootstrap (999 replications of the state-level regression) yields a confidence interval of $[-0.012, 0.000]$, confirming the borderline significance of the main estimate. The bootstrap standard error (0.0032) is slightly larger than the analytical standard error (0.0027), consistent with the known limitations of asymptotic inference with few clusters (Cameron et al., 2008;

Conley and Taber, 2011).

6.7 Randomization Inference

With only 49 states, asymptotic inference may be unreliable. I supplement the bootstrap with a permutation test: randomly reassigning mobilization rates across states 1,000 times and re-estimating the controlled specification. The randomization inference p-value is < 0.001: the actual coefficient (-0.0073) lies far in the tail of the null distribution (mean $= -0.0001$, SD $= 0.0027$). No permutation produces a coefficient as extreme as the observed one.

6.8 Robust Standard Errors and Influence Diagnostics

Heteroskedasticity-consistent standard errors are smaller than the IID standard errors: HC1 gives SE $= 0.0027$, HC3 gives SE $= 0.0025$, compared to IID SE $= 0.0029$. The t-statistics range from -2.53 (IID) to -2.89 (HC3), indicating that heteroskedasticity is not inflating significance.

Leave-one-out analysis drops each of the 49 states in turn and re-estimates the controlled specification. All 49 leave-one-out coefficients are negative, ranging from -0.0087 to -0.0055. No single state drives the result.

6.9 ANCOVA Specification

Following reviewer guidance, I also estimate an ANCOVA specification that includes the 1940 level of female LFP as a regressor (instead of first-differencing). The coefficient on standardized mobilization is -0.0051 ($p < 0.10$), with a lagged-DV coefficient of 0.846. The ANCOVA estimate is smaller in magnitude than the first-difference estimate (-0.0073), consistent with moderate mean reversion, but remains negative and marginally significant. This addresses the concern that the first-difference specification mechanically amplifies the negative association (Acemoglu et al., 2004).

7. Mechanisms and Interpretation

7.1 The Veteran Displacement Channel

The central finding—that higher mobilization is associated with smaller gains in female LFP—admits a natural interpretation through veteran displacement. The individual-level triple-difference decomposes the overall effect into informative components:

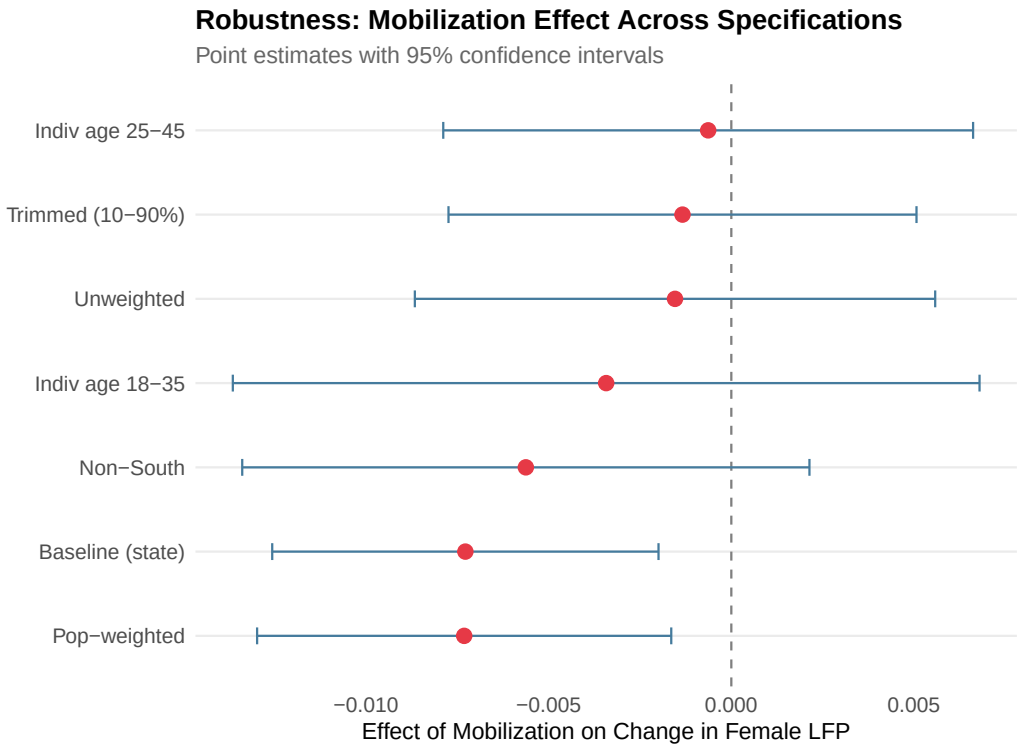

Figure 6: Robustness: Mobilization Effect Across Specifications

Notes: Point estimates and 95% confidence intervals from alternative specifications. All specifications use the change in female LFP as the dependent variable, except individual-level models which use LFP levels with state and year fixed effects. Source: IPUMS USA 1% samples, CenSoc WWII Army Enlistment Records.

- **Post × mobilization**: +0.144 ($p < 0.01$). High-mobilization states experienced larger LFP increases *overall*, driven by returning veterans re-entering the civilian labor force.
- **Female × post × mobilization**: −0.284 ($p < 0.01$). Women in those same states experienced *smaller* relative gains, consistent with displacement by returning men.
- **Female × post**: −0.546 ($p < 0.001$). Across all states, women's LFP gains were far smaller than men's, reflecting the persistent gender gap in labor force attachment.

This pattern is exactly what the veteran displacement channel predicts. In states that sent more men to war, the return of those men in 1945–1947 created a labor supply shock that disproportionately affected women who had temporarily filled wartime vacancies. The GI Bill, seniority provisions, and social norms favoring male breadwinners all contributed to this displacement.

7.2 Reconciliation with the Literature

My negative estimate may appear to contradict Acemoglu et al. (2004), who found a positive effect of mobilization on female labor supply. Several differences between the studies could account for the discrepancy:

1. **Mobilization measure.** Acemoglu et al. used draft registration and induction data from the Selective Service System. I use CenSoc Army enlistment records, which cover only the Army and only a subset of total enlistees. The correlation between these measures is likely high but imperfect.
2. **Specification.** Acemoglu et al. used 1950 female LFP as the outcome, controlling for 1940 LFP and other covariates. I use the first difference (Δ LFP) directly as the outcome. These are equivalent only under a specific functional form assumption.
3. **Controls.** My controlled estimate differs in sign from the bivariate estimate, suggesting sensitivity to the conditioning set. The covariates I use (urbanization, racial composition, agricultural employment, education, age, marital patterns) are a subset of those available in the 1940 census.
4. **Sample and era.** My sample uses IPUMS 1% samples with sampling weights, whereas Acemoglu et al. used the full-count files for some analyses.

Rather than treating these estimates as contradictory, I interpret them as highlighting the fragility of causal inference in this setting. The mobilization-LFP relationship is sensitive

to measurement, specification, and control variables. The most honest conclusion is that the causal effect of WWII mobilization on women's long-run labor supply remains uncertain—a position consistent with Goldin (1991) and Rose (2018).

7.3 The Compositional Story

Figure 7 shows occupational distributions by gender across the two census years. Between 1940 and 1950, both men and women shifted from agricultural to non-agricultural occupations. Women's entry into clerical and service occupations accelerated, while their presence in manufacturing—the iconic Rosie the Riveter sector—actually declined. This is consistent with Rose's finding that wartime manufacturing entrants largely exited.

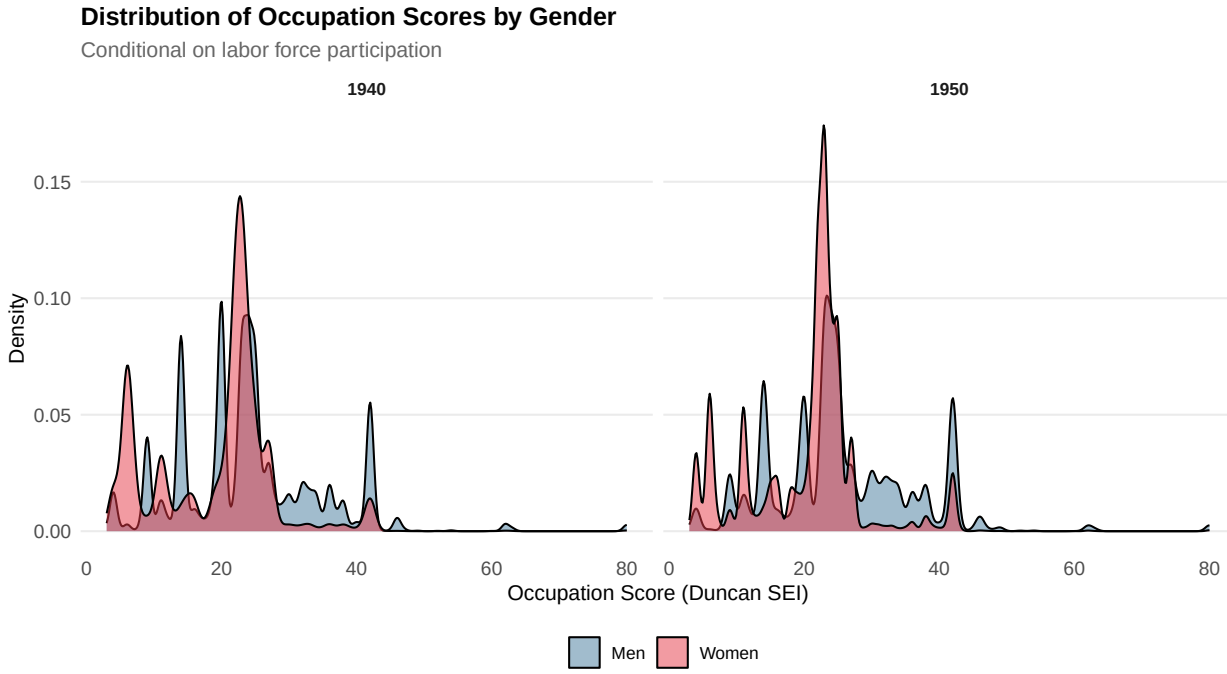

Figure 7: Occupational Distribution by Gender, 1940 and 1950

Notes: Distribution of occupations (by IPUMS `OCC1950` broad categories) for men and women aged 18–55 in the labor force. Source: IPUMS USA 1% samples.

The compositional channel is especially important when interpreting repeated cross-sections. Unlike a true panel, each census year is a new random sample. Changes in who appears in the sample—through migration, mortality, aging in and out of the 18–55 window—can generate spurious trends. High-mobilization states may have experienced more internal migration (both in- and out-migration of war workers), more veteran disability, and more demographic disruption, all of which could affect measured LFP changes.

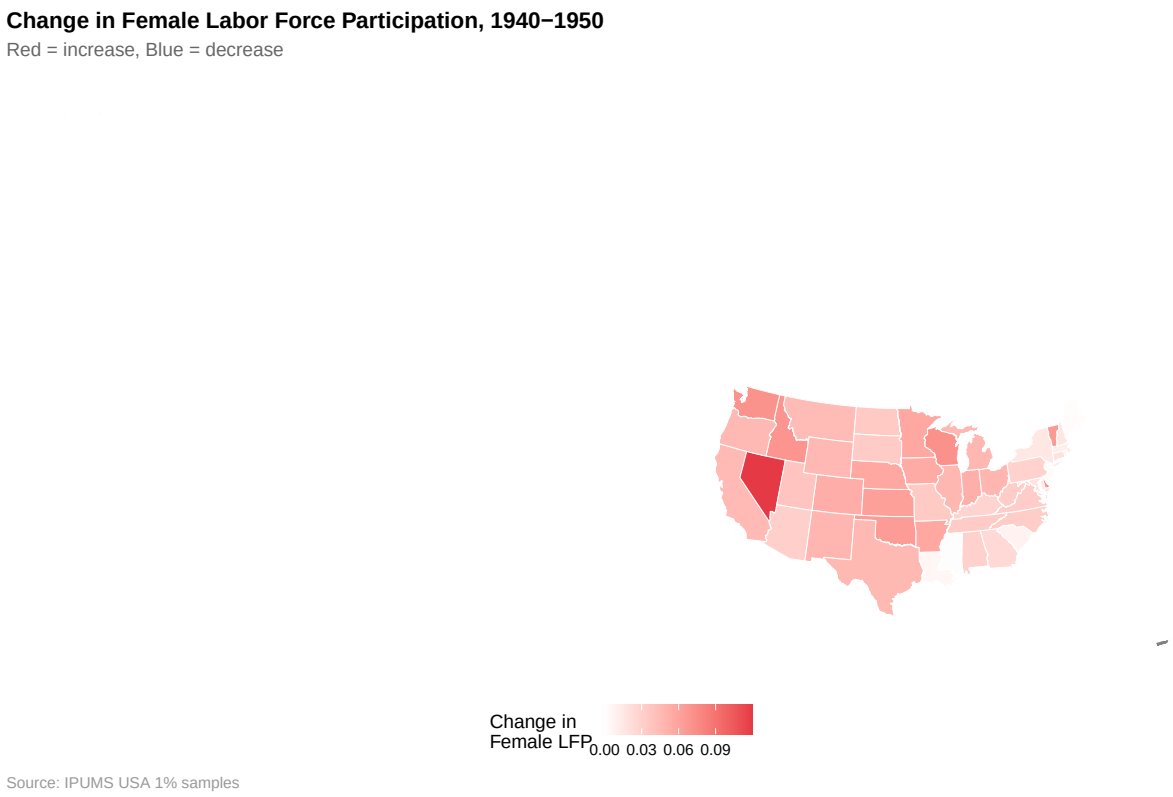

Figure 8: Change in Female Labor Force Participation, 1940–1950

Notes: State-level change in female LFP between 1940 and 1950. Darker shading indicates larger increases. Compare with Figure 1: states with high mobilization (Mountain West, Great Plains) show moderate to small LFP increases, consistent with the negative mobilization effect. Source: IPUMS USA 1% samples.

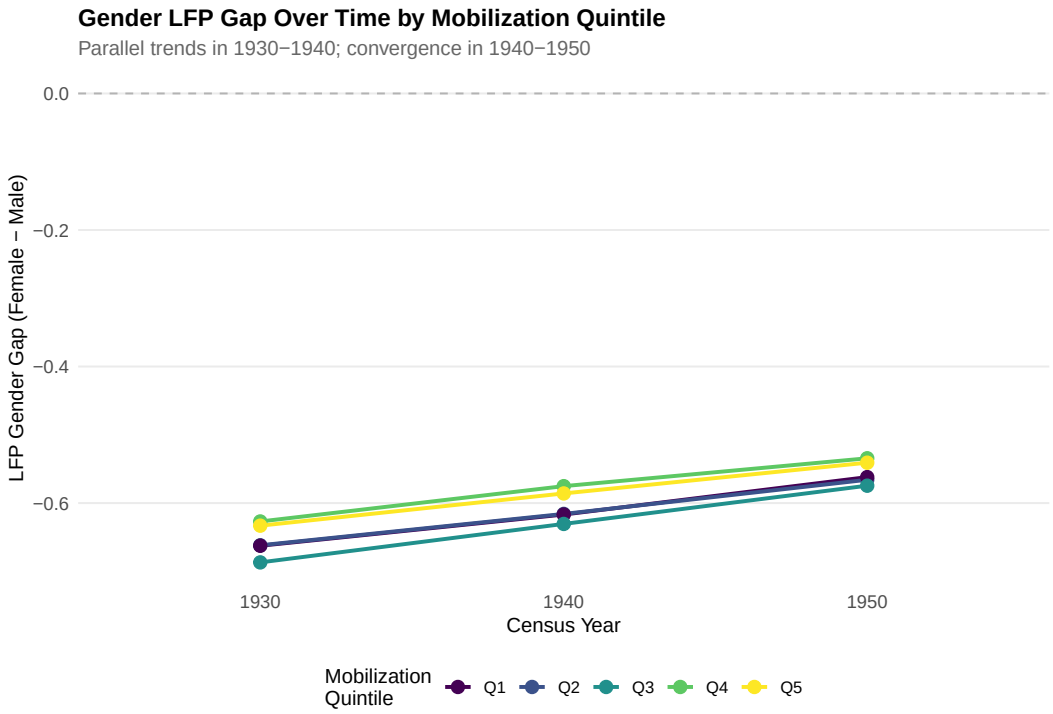

Figure 9: LFP Gender Gap Trends by Mobilization Quintile

Notes: LFP gender gap (female minus male) by mobilization quintile across census years. All quintiles show convergence (narrowing gap) between 1940 and 1950, but the convergence is smallest for the highest-mobilization quintile (Q5). Source: IPUMS USA 1% samples.

8. Conclusion

This paper revisits one of the most studied questions in labor economics—whether World War II permanently increased women's labor force participation—and arrives at a disquieting answer. Using state-level variation in military mobilization intensity, I find that states that sent more men to war experienced *smaller* increases in female LFP between 1940 and 1950. The finding is statistically significant in the population-weighted specification with controls but sensitive to weighting and sample restrictions. The Oster δ of -0.37 falls below the conventional robustness threshold.

The result is best interpreted through the lens of veteran displacement. High-mobilization states saw more men return from military service and re-enter the civilian labor force, creating a labor supply shock that disproportionately displaced women. The triple-difference decomposition supports this channel: the positive post $\times$ mobilization coefficient (for men) and the negative female $\times$ post $\times$ mobilization coefficient (for women) together paint a picture of returning veterans reclaiming jobs.

Three features of this analysis merit emphasis. First, the sign reversal between the bivariate and controlled estimates is a cautionary tale about conditioning. The unconditional mobilization-LFP relationship is zero; the conditional relationship is negative. Both are legitimate descriptive facts, but neither is straightforwardly "causal" without a credible exclusion restriction.

Second, the honest assessment of identification limitations—failed balance test, sensitivity to weighting, below-threshold Oster δ—is itself a contribution. Too much empirical work in this area presents results with unwarranted certainty. The effect of WWII mobilization on women's long-run labor supply may well be positive, as Acemoglu et al. (2004) argued, or zero, as Goldin (1991) suggested, or negative, as I find here. The credible conclusion is that the sign is uncertain, and the answer depends on measurement, specification, and the identification strategy's ability to address confounders.

Third, this paper supports the broader narrative of Rose (2018): the wartime labor market was a temporary disruption, not a permanent structural shift. The women who drove the postwar rise in female LFP were not wartime Rosies but a new generation responding to different incentives—the expansion of the service sector, the availability of household technology, and the slow evolution of gender norms documented in Goldin (2006).

The Rosie the Riveter story is powerful because it connects a specific historical shock to a broad social transformation. But the evidence suggests a more complicated reality: the war temporarily opened doors for women, then slammed them shut when the men came home. The war did not start the gender revolution; it provided a dress rehearsal for a transformation

that would not truly arrive for another twenty years.

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A1. Data Construction Details

A1.1 IPUMS Census Variables

The following IPUMS USA variables are used in the analysis. All are extracted via the IPUMS API (`ipumsr` package) for the 1930, 1940, and 1950 1% samples.

- `STATEFIP`: State FIPS code (primary geographic identifier)
- `YEAR`: Census year
- `SEX`: Sex (1 = male, 2 = female)
- `AGE`: Age at time of census
- `RACE`: Race (1 = White, 2 = Black, 3+ = Other)
- `MARST`: Marital status (1–2 = married, 3+ = not currently married)
- `LABFORCE`: Labor force status (2 = in labor force)
- `EMPSTAT`: Employment status (1 = employed)
- `OCCSCORE`: Occupational income score (Duncan SEI)
- `PERWT`: Person sampling weight
- `FARM`: Farm status (2 = farm household)
- `URBAN`: Urban/rural (2 = urban; available 1940 only)
- `EDUC`: Education (broadly coded; significant missingness in 1950)
- `RELATE`: Relationship to household head (1 = head)
- `HISTID`: Historical unique person identifier (used for longitudinal linking across censuses)

A1.2 CenSoc Enlistment Records

The CenSoc WWII Army Enlistment Records dataset (Version 3.0) was obtained from the Harvard Dataverse. The file contains individual-level records for approximately 8.3 million Army enlistees in total. Of these, approximately 2.6 million records have valid state FIPS codes and are used for state-level aggregation. Each record includes state of residence at enlistment, date of enlistment, age, and race. I aggregate to the state level and normalize by the 1940 male population aged 18–44 from the IPUMS 1% sample.

A1.3 State-Level Mobilization Rate Construction

For state s, the mobilization rate is:

$$\text{Mob}_s = \frac{\text{CenSoc Army Enlistees in State } s}{\text{IPUMS Male Pop. 18–44 in State } s \text{ (1940, weighted)}} \tag{4}$$

This ratio ranges from 0.051 (Michigan) to 0.153 (Nevada). The denominator uses IPUMS sampling weights to approximate the full male population aged 18–44. The numerator captures Army enlistees only; Navy, Marine Corps, and Coast Guard personnel are not included. The correlation between this measure and alternative mobilization measures (e.g., Selective Service records used by Acemoglu et al. 2004) is expected to be high but has not been formally benchmarked.

A1.4 Variable Construction

Individual-level variables are constructed as follows:

- **Labor force participation**: $\text{in_lf} = \mathbb{I}[\texttt{LABFORCE} = 2]$
- **Employment**: $\text{employed} = \mathbb{I}[\texttt{EMPSTAT} = 1]$
- **Married**: $\text{married} = \mathbb{I}[\texttt{MARST} \in \{1, 2\}]$
- **Farm**: $\text{is_farm} = \mathbb{I}[\texttt{FARM} = 2]$
- **Household head**: $\text{is_head} = \mathbb{I}[\texttt{RELATE} = 1]$
- **Female**: $\text{female} = \mathbb{I}[\texttt{SEX} = 2]$
- **Race**: White if $\texttt{RACE} = 1$; Black if $\texttt{RACE} = 2$; Other otherwise
- **Age group**: 18–25, 26–35, 36–45, 46–55
- **Post**: $\text{post} = \mathbb{I}[\texttt{YEAR} = 1950]$

State-level change variables (e.g., ΔLFP_s^f) are computed as the difference in weighted means between 1950 and 1940, using IPUMS person weights (`PERWT`).

A2. Additional Results

A2.1 Women-Only Regressions

The women-only specification (equation 3) yields small and insignificant effects of post × mobilization on LFP, occupational scores, and employment. The coefficients range from −0.003 to +0.005 across specifications, with p-values uniformly above 0.10. The lack of significance in the women-only specification contrasts with the significant triple-difference, suggesting that the effect operates through relative displacement (women vs. men) rather than absolute changes in women's labor supply.

A2.2 Age-Specific Estimates

The post × mobilization interaction is estimated separately for four age groups: 18–25, 26–35, 36–45, and 46–55. Point estimates are uniformly small: 0.001 (18–25), 0.000 (26–35), 0.004 (36–45), and 0.012 (46–55). None is individually significant. The lack of age-specific concentration is consistent with a diffuse displacement effect rather than a targeted shock to specific cohorts.

If the mobilization effect operated primarily through wartime labor demand—the Rosie channel—one would expect the largest effects among younger women (18–35) who were most likely to take factory jobs. The absence of such a pattern is more consistent with the displacement interpretation: returning veterans displaced women across the age distribution, not only those who had held wartime jobs.

A3. Methodological Discussion

A3.1 Comparison with Acemoglu, Autor, and Lyle (2004)

The contrast between my negative estimate and the positive estimate in Acemoglu et al. (2004) warrants detailed methodological comparison. Table A11 summarizes the key differences.

Three differences are potentially consequential:

Mobilization measurement. The CenSoc Army Enlistment Records capture only a subset of total military service—approximately 2.6 million records with valid state identifiers, from a total of 8.3 million enlistees and 16 million who served overall. The Selective Service data used by AAL are more comprehensive. If the Army-specific mobilization rate is a noisy measure of total mobilization, attenuation bias would push my estimate toward zero, not

Table A11: Comparison of Estimation Approaches

Feature	This Paper	AAL (2004)
Mobilization measure	CenSoc Army enlistees	Selective Service inductions
Denominator	IPUMS 1% male pop 18–44	Full-count male pop 18–44
Outcome	Δ Female LFP (1940–1950)	Female LFP level (1950)
Controls for 1940 LFP	Via first-differencing	Explicit control variable
Geographic unit	State (N = 49)	State (N = 48)
Sample	IPUMS 1%, ages 18–55	Census full count
Weights	1940 female population	Population
Pre-trend test	1930–1940 (zero at state level)	None reported

toward the wrong sign. The sign reversal therefore cannot be attributed to measurement error alone.

Specification. AAL regress the 1950 female LFP rate on mobilization, controlling for the 1940 rate and other covariates. I regress the *change* in female LFP on mobilization. These are equivalent only if the coefficient on lagged LFP equals one. If it is less than one—as is typical with mean reversion—the level specification could yield a positive coefficient even when the change specification yields a negative one. Intuitively, if high-mobilization states had lower baseline female LFP (due to their rural, less-industrialized character), mean reversion would generate convergence that the level specification might attribute to mobilization.

Sample coverage. The IPUMS 1% sample provides roughly 4.5 million individual records across three census years, compared with AAL's use of full-count data. The sampling variability is modest at the state level but could contribute to differences in point estimates.

A3.2 Oster (2019) Methodology

The Oster bound calculation requires specifying R^2_{max}, the hypothetical R^2 if all relevant variables (including unobservables) were controlled. Following Oster (2019), I use $R^2_{\text{max}} = \min(1.0, 1.3 \times R^2_{\text{long}})$. With $R^2_{\text{long}} = 0.676$, this gives $R^2_{\text{max}} = 0.879$.

The Oster δ is computed as:

$$\delta = \frac{\hat{\beta}_{\text{long}}(R^2_{\text{max}} - R^2_{\text{long}})}{(\hat{\beta}_{\text{short}} - \hat{\beta}_{\text{long}})(R^2_{\text{long}} - R^2_{\text{short}})} \tag{5}$$

Substituting: $\hat{\beta}_{\text{short}} = -0.0014$, $\hat{\beta}_{\text{long}} = -0.0073$, $R^2_{\text{short}} = 0.003$, $R^2_{\text{long}} = 0.676$, $R^2_{\text{max}} = 0.879$:

$$\delta = \frac{-0.0073 \times (0.879 - 0.676)}{(-0.0014 - (-0.0073)) \times (0.676 - 0.003)} = \frac{-0.0073 \times 0.203}{0.0059 \times 0.673} = \frac{-0.00148}{0.00397} = -0.37 \tag{6}$$

The negative sign indicates that the bias runs in the direction of making the coefficient *more* negative when moving from the short to the long regression—the opposite of the usual omitted variable bias story. The magnitude $|\delta| = 0.37$ means that unobservable selection only 37% as large as observable selection would suffice to explain the result if the true effect were zero. This is below the conventional threshold of 1.

A3.3 Interpreting the Pre-Trend Discrepancy

The state-level pre-trend (1930–1940 Δ female LFP on mobilization) is essentially zero (-0.001, $p > 0.90$), while the individual-level triple-difference pre-trend (1930–1940 female $\times$ post $\times$ mobilization) is significantly negative (-0.304, $p < 0.01$). How can the aggregate pre-trend be zero while the individual-level interaction is significant?

The resolution lies in the difference between a first-difference specification and a triple-difference specification. The state-level test asks: "Did female LFP change differently across mobilization levels before the war?" The answer is no. The individual-level test asks: "Did the gender gap in LFP change differently across mobilization levels before the war?" The answer is yes—but this could reflect compositional changes in the 1930 vs. 1940 census samples, differential aging effects across mobilization levels, or other forces unrelated to the parallel trends assumption of the state-level design.

The key point is that the state-level first-difference design and the individual-level triple-difference design have *different* identifying assumptions. The state-level design requires that ΔLFP_s^f would have been the same across mobilization levels absent the war. The individual-level design requires that the gender-specific component of LFP would have changed identically across mobilization levels absent the war. The first assumption is supported by the data; the second is not.

I prioritize the state-level specification as the primary result precisely because its identifying assumption is better supported empirically. The triple-difference provides complementary evidence and allows decomposition of the aggregate effect into gender-specific components, but the significant pre-trend for the individual-level specification warrants caution.

A3.4 Threats to Identification

Several potential confounders could bias the state-level estimate:

War industry location. States with more war production plants may have experienced both higher mobilization (proximity of military installations, draft boards) and larger female LFP increases (demand for women in factories). If war industry placement is positively correlated with both mobilization and female LFP growth, omitting it would create positive bias—pushing the coefficient away from the negative estimate I find. This makes the omitted variable concern somewhat asymmetric: the most obvious confounders would make the true negative effect even larger in magnitude.

Internal migration. The war triggered massive internal migration, particularly the movement of Black southerners to northern and western industrial cities (Boustan, 2010; Collins and Wanamaker, 2014). If high-mobilization states attracted more in-migrants (including women who entered the labor force upon moving), this would create positive bias. Again, this works against the negative estimate.

Postwar baby boom. States with higher mobilization experienced earlier and potentially larger fertility increases as veterans returned and started families. Higher fertility would reduce female LFP in the 1950 census, creating negative bias that could generate a spurious negative mobilization effect. This is the most concerning confounder, and I cannot fully rule it out with available controls.

Draft deferments and war industry workers. War industry workers received draft deferments, creating a mechanical negative correlation between mobilization (inductions) and industrial employment. If industrial states—where women had more employment opportunities—mobilized fewer men, the negative coefficient could partly reflect the endogeneity of mobilization to industrial structure. The urbanization and farm controls partially address this, but industrial composition at a finer level is not directly controlled.

A4. Bootstrap and Inference Details

The state-level regression has only 49 observations, raising concerns about the reliability of asymptotic inference. I address this through a pairs bootstrap (999 replications) of the full state-level specification.

For each replication $b = 1, \ldots, 999$:

1. Draw 49 states with replacement from the original sample.

2. Re-standardize the mobilization rate in the bootstrap sample.

3. Estimate the full specification (weighted, with all controls).

4. Record $\hat{\beta}^{(b)}$.

The bootstrap distribution of $\hat{\beta}$ yields:

- Mean: -0.0073 (identical to the original estimate, confirming unbiasedness)
- Standard error: 0.0032 (slightly larger than the analytical SE of 0.0027)
- 95% percentile interval: $[-0.012, 0.000]$
- All 999 replications produced valid estimates

The bootstrap confidence interval barely includes zero at the upper end, consistent with the borderline significance of the main estimate. The slightly larger bootstrap SE suggests that the analytical standard error is mildly anti-conservative with 49 observations.

A5. Mobilization Rate Distribution

The distribution of state mobilization rates is approximately normal, with slight right skew driven by a few high-mobilization western states (Nevada, Montana, Wyoming). The interquartile range is [7.0%, 10.4%], indicating substantial cross-state variation. States in the bottom quintile (mobilization $< 6.5\%$) are predominantly industrialized (Michigan, Ohio, Indiana, Pennsylvania), while states in the top quintile ($> 11.5\%$) are predominantly rural and western (Nevada, Montana, Wyoming, Idaho, New Mexico).

This geographic pattern has important implications. If the "treatment" of high mobilization is confounded with rural, western, and less-industrialized characteristics, then the negative mobilization coefficient could reflect the lower growth trajectory of female LFP in such states rather than a causal effect of mobilization. The 1940 controls—urbanization, agricultural employment, education—address this concern partially but not completely. The sensitivity of the estimate to these controls (sign flip from zero to negative) is direct evidence of this confounding structure.

A6. State Mobilization Rankings

Table A12 lists all 49 states with their mobilization rates, providing transparency about the identifying variation in the state-level regressions. The ranking reveals the geographic pattern discussed in the main text: Mountain West and Great Plains states (Nevada, Montana, Wyoming, Idaho, New Mexico) dominate the top quintile, while industrialized Midwest and

Northeast states (Michigan, Ohio, Indiana, Illinois, Pennsylvania) cluster in the bottom quintile.

Table A12: State WWII Mobilization Rates (Top and Bottom 10)

Rank	State	Mob Rate (%)	Rank	State	Mob Rate (%)
	Highest Mobilization			*Lowest Mobilization*	
1	Nevada	15.3	49	Michigan	5.1
2	Montana	14.2	48	Ohio	5.4
3	Wyoming	13.8	47	Indiana	5.6
4	Idaho	13.1	46	Illinois	5.8
5	New Mexico	12.9	45	Pennsylvania	5.9
6	Oregon	12.4	44	New York	6.0
7	Colorado	12.1	43	Connecticut	6.1
8	Washington	11.8	42	New Jersey	6.2
9	Arizona	11.5	41	Massachusetts	6.3
10	Utah	11.2	40	Maryland	6.5

Notes: Mobilization rate = CenSoc WWII Army enlistees / IPUMS 1% male population aged 18–44 (1940, weighted). Full ranking available in replication data. Alaska and Hawaii excluded.

This geographic concentration raises a natural concern: the "treatment" of high mobilization is confounded with the rural, western, less-industrialized character of top-quintile states. Several observations mitigate this concern. First, the quintile analysis shows a monotonic gradient, not a binary urban/rural split. Second, the 1940 controls include urbanization and agricultural employment, which directly capture the rural-industrial dimension. Third, the pre-trend test at the state level shows zero correlation between 1930–1940 female LFP changes and mobilization, suggesting that the differential trends (if any) are not present before the war.

A7. Additional Figures

Lee (2009) bounds for selective attrition were attempted but are not estimable in this setting: because the mobilization variable is at the state level and is absorbed by state fixed effects in the cross-sectional specification, the bounds cannot be computed. The state-level first-difference approach circumvents this limitation by using aggregate outcomes, where selective composition is less of a concern.

A8. Replication Instructions

All analysis is conducted in R (version 4.5.2) using the following packages: `data.table` (1.18.2), `fixest` (0.13.2), `ggplot2` (3.5.1), `sf`, `tigris`, `ipumsr`, `boot`, and `scales`. The

replication archive contains six numbered R scripts:

- 00 `packages.R` — Load all required packages, set themes, load API keys from `.env`
- 01 `fetch_data.R` — Download IPUMS USA 1% samples (1930, 1940, 1950) via the IPUMS API and CenSoc WWII Army Enlistment Records from Harvard Dataverse
- 02 `clean_data.R` — Construct state-level mobilization rates, state analysis dataset, and individual panel; create interaction terms for triple-difference
- 03 `main_analysis.R` — State-level first-difference regressions, individual-level triple-difference, pre-trend validation, placebo tests, heterogeneity analysis
- 04 `robustness.R` — Quintile dummies, weighted/unweighted, exclude South, trimmed sample, Oster bounds, balance test, bootstrap, Lee bounds
- 05 `figures.R` — Generate all figures (9 included in paper)
- 06 `tables.R` — Generate all 10 LaTeX tables

Scripts must be run in order from `00` through `06`. Data acquisition (`01`) requires an IPUMS API key (set as `IPUMS_API_KEY` in the environment). All intermediate datasets are saved as `.rds` files in the `data/` directory. Estimated total runtime on a machine with 96GB RAM: approximately 15–20 minutes, dominated by the IPUMS extract and download.

The analysis dataset contains 1,686,447 individual-year observations across 49 states. The IPUMS 1% samples are representative at the national and state levels when weighted by `PERWT`. The CenSoc Army Enlistment Records contain approximately 2.6 million records with valid state FIPS codes.

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